

User interfaces

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User interface definition

A key concept when discussing the accessibility and usability of ICT products and services concerns the user interface. This part of the system allows a person to interact with it. Its main purpose involves the following:

- The user enters commands into the system so that it performs a specific action (input).
- The system informs users of its status and any changes to it (output).

User interface components

Formally speaking, the following components can be distinguished in communication between a person and a system:

- People have channels of perception (tactile, auditory, visual) and channels of expression (speech, movement), all of which are connected to the human brain's system of representation.
- The system's user interface receives input from human expression channels and sends output to human perception channels. System input/output channels are controlled by a processor, which in turn communicates with the rest of the system via data transmission and reception components.

User interfaces generally have two main parts, which define their scope in relation to the user and the system itself:

- The Sensory Interface is the physical point of contact between the user and the system. It defines the forms (visual, auditory, tactile) in which the user receives and sends information to and from the system. It is made up of input and output channels. Typical components of a Sensory Interface include:
 - Input channels: Keyboard, Mouse or Mouse Emulator, Joystick, Voice Recognition, etc.
 - Output channels: Screen, Speakers, Headphones, Screen Reader, etc.
- The Application Interface acts as a boundary between the user interface and the rest of the system. It is made up of the processing module and the system's data transmission and reception modules. It performs two main operations:
 - Defining and managing user-system communication processes.
 - Linking the User Interface with the rest of the system.

[Figure 3.1](#) below illustrates the main components involved in user-ICT-based system communication.

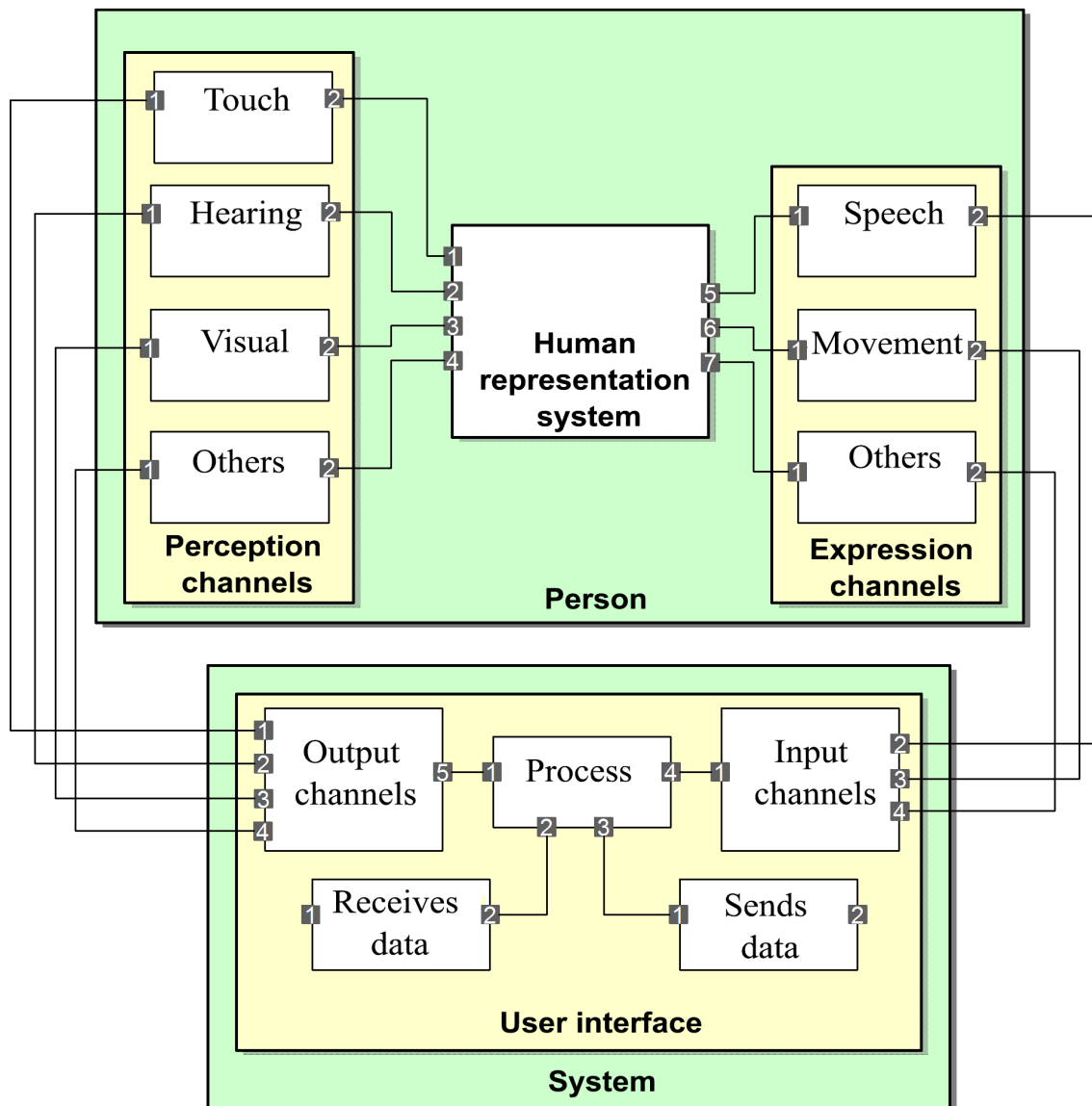


Figure 3.1. Interface model

The user interface model suggests that individuals process information through channels like touch, hearing, and sight, and respond through speech or movement. All these channels link together within the human representational system. A system's user interface has input channels, connected to the channels of human expression, and output channels, connected to those of human perception. System input/output channels are controlled by a processor, which in turn communicates with the rest of the system via data transmission and reception components.

A very brief overview of user interface accessibility

This section is a very brief account of the user interfaces that PCs have offered over time, along with the accessibility solutions adopted for a particular group of users: visually impaired people.

Initially, information was entered via a keyboard and displayed as a series of command lines on the screen. At that time, a device called the Optacon was

invented. It enabled a visually impaired person to read information from the screen, by converting the text displayed on the screen or on a page of a book into Braille, line by line.

Not long after, the WIMP (Windows, Icons, Menus and Pointer) paradigm emerged. This screen layout is familiar to all of us. However, this new layout no longer displays information line by line, but organises it across a desktop and various windows. The solution offered by Optacon for reading line by line was no longer useful. In addition, a new interaction tool appeared in the form of the mouse, a device that is useless to blind people because it requires hand-eye coordination. This is where screen readers came in. These systems interface with the operating system to relay information regarding windows, menus, and icons to visually impaired users through voice or Braille output. Users can operate the computer using a keyboard equipped with touch-sensitive keys specifically designed for accessibility, thereby eliminating the need for a mouse. It is likely that computers will eventually move away from the WIMP paradigm. At some point, visual information will no longer be displayed on a screen but will be shown to us via glasses, or directly onto the iris. The mouse and keyboard will be replaced by our hand gestures, often performed using everyday objects. What challenges will this new paradigm entail for a visually impaired person? This series of events related to innovation and accessibility suggest an endless cycle:

1. Innovative interaction technology emerges.
2. A solution for accessibility must be invented.

When asked whether this is a vicious or virtuous cycle, there are no reliable answers. This topic will be covered in more detail later.